7.8 Four Squares

WORKSHEET

The question

STEP III 1994, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Recall the two-square version: $(a^2+b^2)(p^2+q^2)$ is a sum of two squares because $|z|^2|w|^2 = |zw|^2$ for complex numbers $z = a+b\mathrm{i}$, $w = p+q\mathrm{i}$. How could matrices extend this to four squares?

STAGE 1 The determinant is a sum of four squares

Ask yourself What is the determinant of $\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}$? If $z_1 = a+b\mathrm{i}$ and $z_2 = c+d\mathrm{i}$, what is it in terms of a, b, c, d?

STAGE 3 Reading off the four squares

Pause and try Expand $w_1 = (a+b\mathrm{i})(p+q\mathrm{i}) - (c+d\mathrm{i})(r-s\mathrm{i})$ and $w_2 = (a+b\mathrm{i})(r+s\mathrm{i}) + (c+d\mathrm{i})(p-q\mathrm{i})$ into real and imaginary parts.

What's really going on

Lagrange's four-square theorem

Here is the shape of Lagrange's proof. By Euler's identity, it is enough to show every prime p is a sum of four squares. A counting argument shows that some multiple mp, with $m < p$, is a sum of four squares. A clever "descent" then shows that if mp is a sum of four squares with $1 < m < p$, so is $m'p$ for some smaller m'; repeating, we reach $m = 1$. (Compare Fermat's descent in Section 1.5.) In 1834 Jacobi went much further and found the exact number of ways to write n as a sum of four squares: 8 times the sum of the divisors of n that are not multiples of 4. For example, 1 can be written in 8 ways (± 1 in any of four positions).

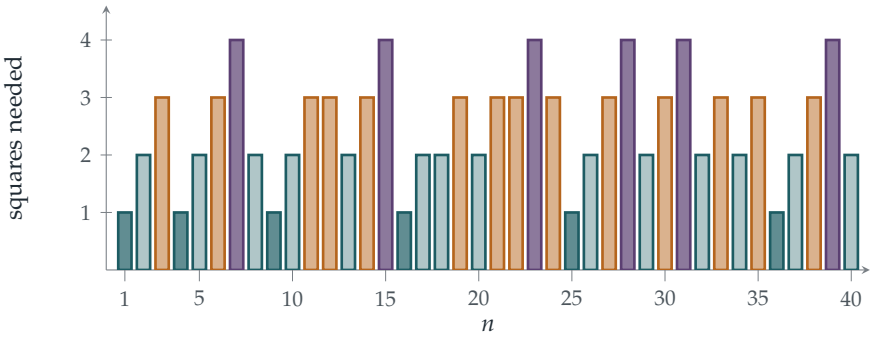


Figure 7.8.1 The least number of squares needed to make each n from 1 to 40. Every number needs at most four, by Lagrange's theorem. The numbers that need four (plum) are $7, 15, 23, 28, 31, 39, \ldots$: exactly those of the form $4^k(8m+7)$, by a theorem of Legendre and Gauss.

Two, three, four

The pattern across dimensions is subtle. Sums of two squares come from complex numbers, and a number is a sum of two squares exactly when its primes of the form $4k+3$ appear to even powers (Section 1.6). Sums of four squares come from quaternions, and every number is one. But sums of three squares have no multiplication identity at all (there is no three-dimensional number system with a multiplicative size; that is why Hamilton failed with triples, Section 6.7), and the theorem is harder: Legendre and Gauss showed that n is a sum of three squares unless $n = 4^k(8m+7)$ (Figure 7.8.1). Beyond four, the eight-dimensional octonions give an eight-square identity, found by Degen and Graves, and Hurwitz proved in 1898 that identities of this kind exist only for 1, 2, 4 and 8 squares.

Waring's problem

Lagrange's theorem is the first case of a question asked by Edward Waring in 1770: is there, for each k, a number $g(k)$ such that every positive integer is a sum of at most $g(k)$ kth powers? David Hilbert proved in 1909 that there is. We now know $g(2) = 4$, $g(3) = 9$ (for example $23 = 2 \cdot 2^3 + 7 \cdot 1^3$ needs nine cubes) and $g(4) = 19$.

Try this next

1. **A worked example.** Use the formulas for $L_1, \dots, L_4$ with $(a,b,c,d) = (p,q,r,s) = (2,1,1,1)$ to write $49 = 7 \times 7$ as a sum of four squares, and check your answer. Is it the same as the obvious $49 = 7^2 + 0^2 + 0^2 + 0^2$?

2. **Two squares inside four.** Show that if $c = d = r = s = 0$, the identity reduces to the two-square identity of Section 1.6.

3. **Numbers needing four squares.** Show that no number of the form $8m + 7$ is a sum of three squares, by considering squares modulo 8.

4. **Jacobi's formula.** Count the representations of 2 as a sum of four squares (with order and signs mattering), and check that the answer is $8 \times (1 + 2) = 24$.

Further reading

- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Chapter V proves Lagrange's four-square theorem.
- **John Conway & Derek Smith, *On Quaternions and Octonions*** (A. K. Peters). The arithmetic of quaternions, including a proof of the four-square theorem via quaternion integers.
- **Aigner & Ziegler, *Proofs from THE BOOK*,** chapter "Representing numbers as sums of two squares". The two-square companion to this section.

The full guided solution is in Section 7.8:
https://beyondstep.keithmcnulty.org/ch07/four-squares.html